CHAPTER 1

Eigenvectors and Eigenvalues

Like many specialized disciplines, Linear Algebra uses many unfamiliar terms whose origins you might wonder about. Eigenvectors and eigenvalues are two of them. If you know German, you will recognize that eigen means inherent or a characteristic attribute. Named by the German mathematician David Hilbert, an eigenvector mathematically describes a characteristic feature of an object that remains unchanged after transformation. You can think of an eigenvector as the direction that doesn not change direction. An eigenvector characterizes a linear transformation, whereas its eigenvalue tells how much the vector is scaled. Eigenvalues can be negative or positive. A negative value indicates the direction of the eigenvector is reversed.

Eigenvalues and eigenvectors are a way to break down matrices, which can simplify many calculations and enable us to understand various properties of the matrix. They are widely used in physics and engineering for stability analysis, vibration analysis, and many other applications.

Let's look at a visual example.

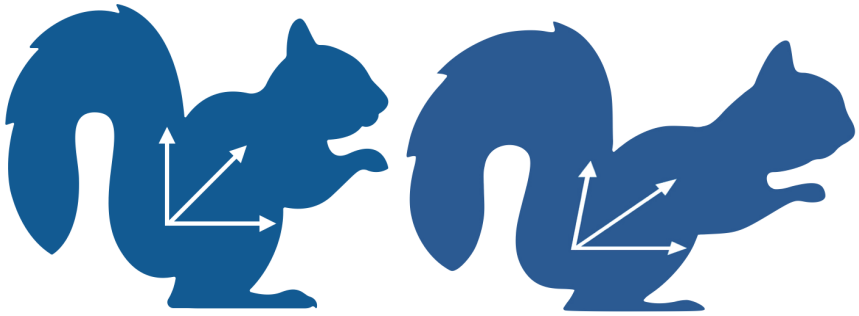


Figure 1.1: Standardized and skewed squirrel image.

You can see that the image on the right is a skewed version of the image on the left. Look closely at the vectors and you will notice that one of the vectors is pointing in the same direction in both images, while the direction of the other two vectors has changed. The eigenvector is the one at the bottom that points to 0 degrees (which you can think of due east) in both images. So, the characteristic attribute of both images is their horizontal direction.

When you overlay the vectors from one image over the other, you will notice that the horizontal vector, while the same direction in both images, is a bit longer in the skewed

version. The scale of the stretch is described by an eigenvalue.

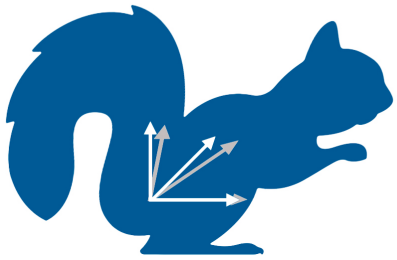


Figure 1.2: Standardized and skewed squirrel image overlaid on each other.

1.1 Definition

Eigenvector and Eigen Equation

Given a square matrix A, a non-zero vector v is an eigenvector of A if multiplying A by v results in a scalar multiple of v. In other words, the *eigenequation* is:

$$Av = \lambda v \tag{1.1}$$

where λ is a scalar known as the eigenvalue corresponding to the eigenvector v.

An eigenvector is a nonzero vector that does not change direction when a linear transformation is applied.

It may stretch, shrink, or flip direction, but it *stays on the same line*.

1.2 Finding Eigenvalues and Eigenvectors

You find the eigenvalues of a matrix A by solving the characteristic equation:

$$\det(A - \lambda I) = 0 \tag{1.2}$$

where $\det(.)$ denotes the determinant, I is the *identity matrix* of the same size as A, and λ is a scalar. In other words, we add λ along the diagonal of a, and solve for its determinant.

$$A - \lambda I = \begin{bmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{bmatrix}$$

Once your find the eigenvalues, you can find the corresponding eigenvectors by substituting each eigenvalue into the equation $Av = \lambda v$, and solving for v. v is a vector which only gets *scaled*, with no change in direction.

For a non-zero **v** to satisfy this equation,

-
- $\det(A - \lambda I) = 0$ must have a determinant of zero
- $A - \lambda I$ cannot have an inverse
- space is flattened on to a volume of 0.

FIXME visual representation graphically?

1.3 Example

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the characteristic equation is:

$$(a - \lambda)(d - \lambda) - bc = 0 \tag{1.3}$$

Solving this equation gives the eigenvalues. Substituting each eigenvalue back into the equation $Av = \lambda v$ gives the corresponding eigenvectors.

Let matrix $A =$

$$\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

The characteristic equation is:

$$|A - \lambda I| = 0$$

$$\begin{bmatrix} 5 - \lambda & 4 \\ 1 & 2 - \lambda \end{bmatrix} = 0$$

$$(5 - \lambda)(2 - \lambda) - (4)(1) = 0$$

$$10 - 5\lambda - 2\lambda + \lambda 2 - 4 = 0$$

$$\lambda 2 - 7\lambda + 6 = 0$$

$$(\lambda - 6)(\lambda - 1) = 0$$

$$\lambda = 6, \lambda = 1$$

Now that you have the eigen values you can substitute these values into the equation:

$$|A - \lambda I| = 0$$

For $= 1$:

$$(A - \lambda I)v = O$$

$$\begin{bmatrix} 5-1 & 4 \\ 1 & 2-1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Next, use elementary row transformation by multiplying row 2 by 4, then subtracting row 1.

$$\begin{bmatrix} 4 & 4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Now you can expand as an equation:

$$4x + 4y = 0$$

Assume $y = w$

$$4x = -4w$$

$$x = -w$$

The solution is:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -w \\ w \end{bmatrix} = w \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

So the eigenvector is:

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Now we need to substitute the other eigenvalue, 6, into the equation and follow the same procedure for finding the eigenvector.

$$\begin{bmatrix} 5-6 & 4 \\ 1 & 2-6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Next, use elementary row transformation by adding row 1 to row 2.

$$\begin{bmatrix} -1 & 4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Expand as an equation:

$$-x + 4y = 0$$

$$\text{Assume } y = w$$

$$-x + 4w = 0$$

$$x = 4w$$

The solution is:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4w \\ w \end{bmatrix} = w \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

So the eigenvector is:

$$\begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

In conclusion, the eigenvectors of the given 2 x 2 matrix are:

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

1.4 Eigenbasis and Eigenspace

Let A be a square matrix and let λ be an eigenvalue of A.

Eigenspace

The **eigenspace** corresponding to λ is the set of all vectors v such that

$$Av = \lambda v.$$

Equivalently, the eigenspace of λ is the solution set of

$$(A - \lambda I)v = 0.$$

Note that there will likely be multiple lambdas for a given matrix, and therefore multiple eigenspaces for each matrix.

The eigenspace is a subspace consisting of all eigenvectors associated with λ, together including the zero vector, $\vec{\mathbf{0}}$.

Eigenbasis

An *eigenbasis* for a matrix A is a basis of the vector space consisting entirely of eigenvectors of A.

$$\{v_1, v_2, \ldots, v_n\}$$

is an eigenbasis for A if:

$$Av_i = \lambda_i v_i \quad \text{for each } i = 1, 2, \ldots, n$$

Exercise 1 A 2×2 matrix

Working Space

Find the eigenbasis of the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$$

Answer on Page 9

1.5 Eigenvalues and Eigenvectors in Python

Create a file called `vectors_eigen.py` and enter this code:

```
# import numpy to perform operations on vector
import numpy as np
from numpy.linalg import eig

a = np.array([[2, 2, 4],
              [1, 3, 5],
              [2, 3, 4]])
eigenvalue,eigenvector = eig(a) #return a tuple of eigenvalues [l1, l2,..., ln],
↪  eigenvectors [ [...], [...], ...]

# The values are not in any particular order
print('Eigenvalues:', eigenvalue)

# The eig function returns the normalize vectors
print('Eigenvectors:', eigenvector)
```

1.6 Connection to transformations

In the previous chapter, we studied matrix transformations such as scaling, reflection, shear, and rotation. Each of these transformations changes vectors in space by stretching, shrinking, flipping, or rotating them. Most vectors change both their length and/or their direction under such transformations.

Eigenvectors arise naturally when a transformation leaves at least one direction unchanged. Different types of transformations exhibit different eigenvector behavior:

- **Scaling transformations** preserve all directions. Every nonzero vector is an eigenvector.
- **Reflections** preserve the direction of vectors along the axis of reflection and reverse vectors perpendicular to it. Vectors lying on the axis of reflection are unchanged by the transformation and therefore are eigenvectors with eigenvalue 1. Vectors perpendicular to the axis of reflection are flipped to point in the opposite direction, making them eigenvectors with eigenvalue -1.
- **Shear transformations** preserve exactly one direction, even though most vectors change direction.
- **Rotations in two dimensions** generally preserve no directions, and therefore have no real eigenvectors (except in the trivial cases of $\theta = \frac{\pi}{2}$ or 0).

These observations explain why some matrices have many eigenvectors, some have only a few, and some have none at all. Eigenvectors exist precisely when a transformation has invariant directions.

Understanding eigenvectors as preserved directions allows us to connect algebraic computations to geometric behavior. When a matrix has enough independent eigenvectors, it becomes possible to describe the transformation entirely in terms of simple scaling along these special directions.

1.7 Summary

In summary, lets take a *transformation*:

Something that stretches, squishes, flips, or rotates space. This transformation could be represented by a matrix.

Most vectors in space will get changed in both direction and length when you apply the transformation.

But some special vectors don't change direction at all — they *only* get stretched or shrunk (and maybe flipped).

These special "unchanging-direction" vectors are called eigenvectors. The amount they get stretched (or shrunk) is their eigenvalue.

1.8 Where to Learn More

Watch this video from Khan Academy, *Introduction to Eigenvectors*: https://rb.gy/mse7i

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Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 6)

First, write the $A - \lambda I$ equation

$$\begin{bmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{bmatrix}$$

Our goal is to get the determinant of $A-\lambda I$ to be equal to zero. For a 2×2, the determinant is $ad - bc$:

$$\begin{aligned} (1-\lambda)(-2-\lambda) - 4 &= 0 \\ (-2+2\lambda-\lambda+\lambda^2) - 4 &= 0 \\ \lambda^2+\lambda-6 &= 0 \\ \lambda_1 = 2, \quad \lambda_2 &= -3 \end{aligned}$$

Then we solve for the two eigenvectors, $\mathbf{e}_n$, using $A - \lambda I$

For $\lambda_1 = 2$:

$$\lambda_1 = 2 : A - 2I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}$$

We solve the following using row-column multiplication (rows of numbers and columns of x_i)

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} -x_1 + 2x_2 = 0 \\ 2x_1 - 4x_2 = 0 \end{cases}$$

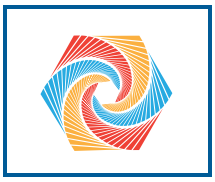
We notice, $x_1 = 2x_2$. This means every eigenvector is a multiple of $\mathbf{e}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ (using $x_2 = 1$) .

For $\lambda_2 = -3$:

$$\lambda_2 = -3 : A + 3I = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} 4x_1 + 2x_2 = 0 \\ 2x_1 + x_2 = 0 \end{cases}$$

This results in $x_2 = -2x_1$, meaning $\mathbf{e}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.



Index

eigenbasis, 6
eigenequation, 2
eigenvalue, 1
    calculating, 2
eigenvector, 1